

Supplementary Table 1. Inter-grader reliability and grader-exclusion sensitivity analysis

A Per-grader reliability vs. each reference standard

Grader	N	Natural · vs Nurse Triage			Multiturn · vs Nurse Triage			Natural · vs Clin-Adjudicated			Multiturn · vs Clin-Adjudicated		
		Agree	Under	κ	Agree	Under	κ	Agree	Under	κ	Agree	Under	κ
1	78	55.1%	35.9%	0.344	69.2%	21.8%	0.496	48.7%	29.5%	0.326	46.2%	23.1%	0.218
2	62	43.5%	37.1%	0.439	40.3%	46.8%	0.459	50.0%	32.3%	0.473	43.5%	43.5%	0.464
3	43	41.9%	53.5%	0.125*	48.8%	44.2%	0.174*	51.2%	30.2%	0.420	51.2%	20.9%	0.356
4	58	65.5%	13.8%	0.514	56.9%	19.0%	0.266	75.9%	12.1%	0.630	58.6%	19.0%	0.372
5	14	64.3%	21.4%	0.395	64.3%	14.3%	0.341	21.4%	57.1%	0.000†	28.6%	50.0%	0.146†

* Weighted κ >1 SD below the group mean vs nurse triage. † Weighted κ >1 SD below the group mean vs clinician-adjudicated.

Group-mean weighted κ — vs nurse triage: Natural 0.363 (SD 0.131), Multiturn 0.347 (SD 0.120); vs clinician-adjudicated: Natural 0.370 (SD 0.210), Multiturn 0.311 (SD 0.114).

Grader 5 is a blind self-reviewer (same person as reviewer; Dataset 1, n=14). Agree = exact agreement; Under = under-triage rate; κ = Cohen's weighted κ. Per-grader mean clinical distance is reported in the analysis workbook.

B Primary-endpoint sensitivity excluding the flagged grader (Grader 3)

Comparison	Agreement full (N=255)	Agreement excl. G3 (N=212)	Under-triage full	P full	Under-triage excl. G3	P excl. G3
Natural vs. Nurse Triage	52.9%	55.2%	70.8% (85/120)	<0.0001	65.3% (62/95)	0.0038
Natural vs. Clinician-Adjudicated	54.1%	54.7%	60.7% (71/117)	0.0261	60.4% (58/96)	0.0519
Multiturn vs. Nurse Triage	55.7%	57.1%	69.0% (78/113)	<0.0001	64.8% (59/91)	0.0061
Multiturn vs. Clinician-Adjudicated	48.2%	47.6%	54.5% (72/132)	0.3384	56.8% (63/111)	0.1837

Under-triage % and its two-sided exact-binomial *P* are computed among disagreement cases. Excluding Grader 3 (N=212) is compared with the full cohort (N=255).